SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 30%

QUESTIONS: 2 **Total Marks:50**

Annexure A

Technical Presentation

Criteria	*Technical Content Accuracy(3)*	*Problem Identification(2.5)*	*Use of Tools(2)*	*Communication(1.5)*	*Professionalism(1)*	*Total(10)*
Weightage	*30%*	25%	20%	20%	10%	*100%*
Q1						
Q2						

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Annexure A

Topic 1: Decision Support Systems (DSS)

🎯 Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

🎯 Focus Areas:

1. OLTP vs OLAP concepts
2. Key differences (data, users, purpose)
3. Performance comparison
4. Use-case scenarios
5. Integration between systems

Topic 3: Architecture of Data Warehouse

🎯 Focus Areas:

1. Data warehouse definition
2. Three-tier architecture
3. ETL process
4. Data marts & metadata
5. Benefits in decision-making

Topic 4: OLAP and Data Cubes

🎯 Focus Areas:

1. OLAP concept
2. Data cube structure
3. OLAP operations (slice, dice, roll-up, drill-down)
4. Applications in analytics

Topic 5: Dimensional Data Modeling

🎯 Focus Areas:

1. Fact and dimension tables
2. Star schema vs snowflake schema
3. Schema design process
4. Real-world example

Topic 6: Query Processing & BI Tools

🎯 Focus Areas:

1. Query processing stages
2. Query optimization
3. BI tools overview
4. Reporting and visualization
5. Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

🎯 Focus Areas:

1. Introduction to KNIME
2. Workflow components (nodes, connections)
3. Performing OLAP operations
4. Sample application/demo

5. Report generation

PART B: PRESENTATION RULES & GUIDELINES

◇ 1. Group Formation

1. Each group: 3–5 students
2. Each member must participate
3. Assign roles:
    1. Presenter
    2. Designer (PPT)
    3. Researcher
    4. Demo/Case handler

◇ 2. Presentation Structure (MANDATORY)

Each group must include:

1. Introduction
    1. Definition and importance
2. Core Concepts
    1. Explanation with diagrams
3. Process/Architecture
    1. Step-by-step explanation
4. Real-World Example
    1. Case study or application
5. Tool/Demo (if applicable)
    1. KNIME or BI tool
6. Conclusion
    1. Summary + future scope

◇ 3. Slide Preparation Rules

1. Maximum slides: 10–15
2. Use:
    1. Diagrams
    2. Flowcharts

    3. Tables (for comparison)
3. Avoid:
    1. Too much text
    2. Copy-paste content

◇ 4. Presentation Delivery Rules

1. Time limit: 10–12 minutes per group
2. Each member must speak at least 2 minutes
3. Maintain:
    1. Eye contact
    2. Clear voice
    3. Logical flow

CSA1607-DATA WAREHOUSING AND DATA MINING

OLAP(ONLINE ANALYTICAL PROCESS)

PRESENTED BY

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WHAT IS OLAP??

OLAP (Online Analytical Processing) is a technology that enables users to analyze large volumes of data quickly from multiple perspectives. It is a key component of Business Intelligence (BI) and Data Warehousing, helping organizations make strategic decisions through fast, interactive, and multidimensional data analysis.

Key Features

- ❑ Fast query processing
- ❑ Multidimensional analysis
- ❑ Interactive reporting
- ❑ Trend and pattern analysis
- ❑ Decision support

"OLAP converts raw data into meaningful business insights."

WHY OLAP IS IMPORTANT??

Organizations generate huge amounts of data every day from sales, customers, finance, marketing, and operations. Traditional databases are designed for transaction processing, but they are not suitable for complex analysis.

OLAP solves this problem by:

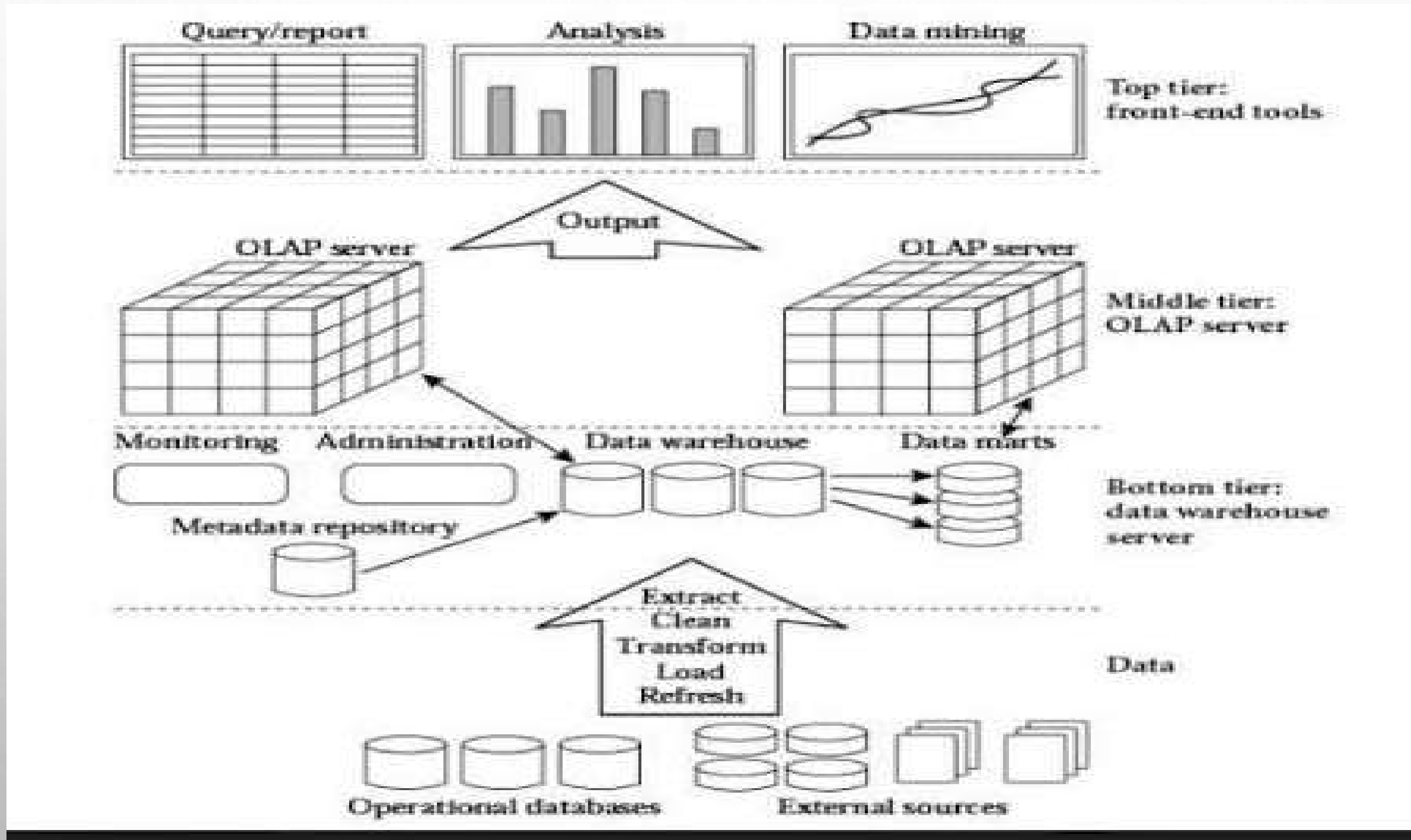
Providing instant access to summarized data.
Comparing business performance across different dimensions.
Identifying trends and hidden patterns.
Supporting strategic planning and forecasting.
Improving business productivity and profitability.

Example

A retail company can instantly compare:

Sales by Product
Sales by Region
Sales by Month
Sales by Customer Category

OLAP ARCHITECTURE



OLAP OPERATIONS

Roll-Up

- Summarizes data.
- Example: Monthly Sales → Yearly Sales.

Drill-Down

- Displays detailed information.
- Example: Year → Month → Day.

Slice

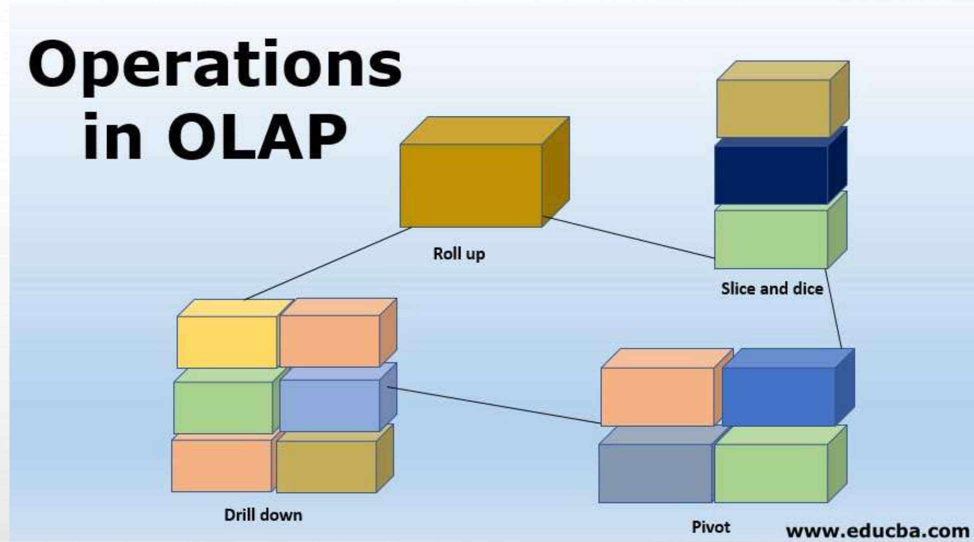
- Analyzes one dimension.
- Example: Sales in Chennai.

Dice

- Filters multiple dimensions.
- Example: Laptop Sales in Chennai during June.

Pivot

- Rotates the data view.
- Example: Compare Regions instead of Products.



TYPES OF OLAP

Types of OLAP

MOLAP (Multidimensional OLAP)

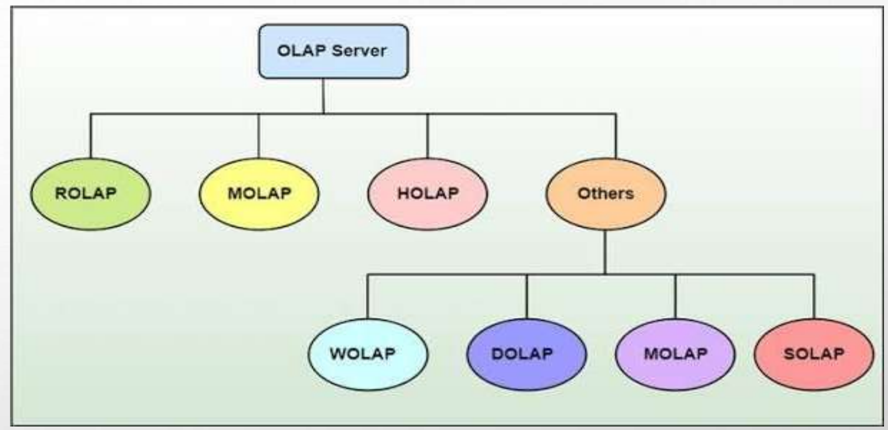
- ❑ Stores data in multidimensional cubes.
- ❑ Very fast query performance.
- ❑ Best for complex analysis.

ROLAP (Relational OLAP)

- ❑ Uses relational databases.
- ❑ Handles very large datasets.
- ❑ More scalable than MOLAP.

HOLAP (Hybrid OLAP)

- ❑ Combines MOLAP and ROLAP.
- ❑ Provides both speed and scalability.



ADVANTAGES,LIMITATIONS AND FUTURE SCOPE

Advantages

- High-speed data analysis
- Interactive reports and dashboards
- Better decision making
- Supports forecasting
- Improves business performance

Limitations

- High implementation cost
- Requires skilled professionals
- Large storage requirements
- Complex maintenance

Future Scope

- Cloud-based OLAP
- AI-powered analytics
- Real-time data analysis
- Predictive Business Intelligence
- Big Data integration